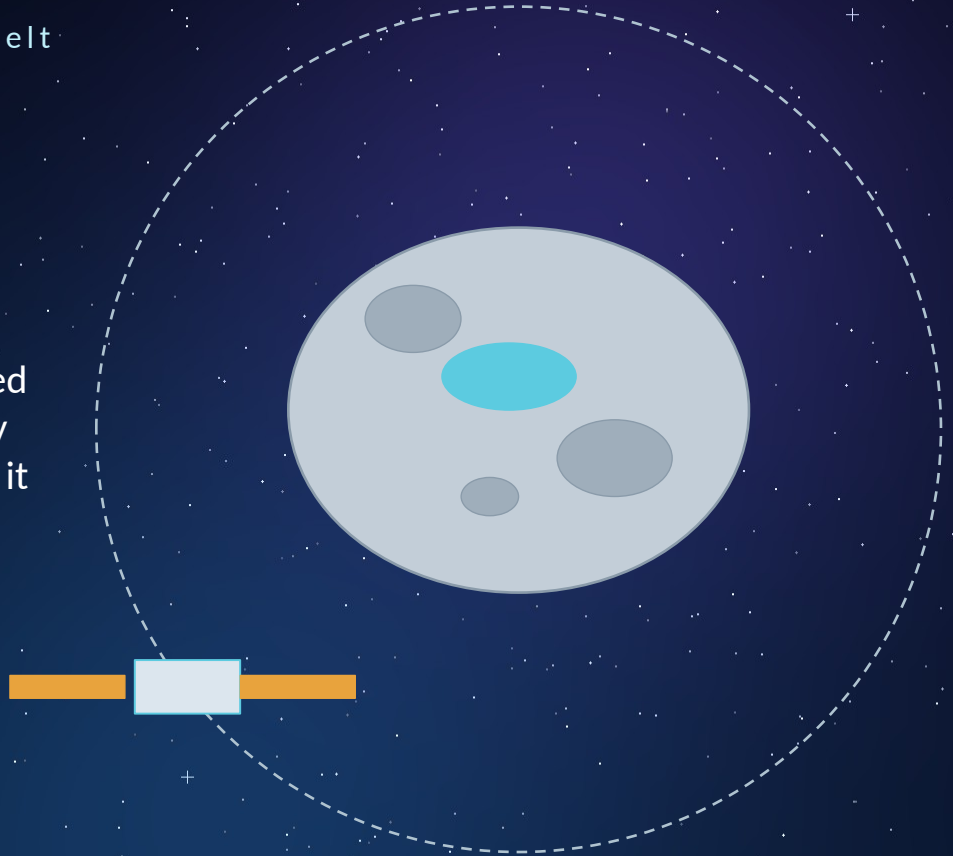
STAC WARS | PS4 | Mining the Asteroid Belt

ICE FORGE

Make asteroid water cheaper than Earth-launched water for customers who are already in space, by processing it inside sealed batch cells and selling it where it is used.



Team: TEAM ROCKET

1 Resource and mission thesis

Water ice is the one asteroid resource whose first customer is already in space

Problem

Every kilogram used in orbit is lifted from Earth first. Water is the heaviest routine consumable: life support, radiation shielding and, after electrolysis, hydrogen/oxygen propellant.

Opportunity

A carbonaceous or icy small body can hold water in the regolith. If it is delivered to a depot beyond low Earth orbit, the customer skips the Earth-to-depot launch, which is the dominant cost there.

Alternative resources compared

Platinum-group metals: returning them needs a re-entry capsule, and any large delivery would crash the Earth price. Rejected as the first product. Water is kept; metals stay a later option.

Product

Purified water (target ≥ ASTM-type "potable" grade, to be confirmed with customer) delivered in insulated tanks to a cislunar depot.

First customer

Operators of propellant depots, stations and tugs beyond LEO. Hypothesis only: no signed demand yet.

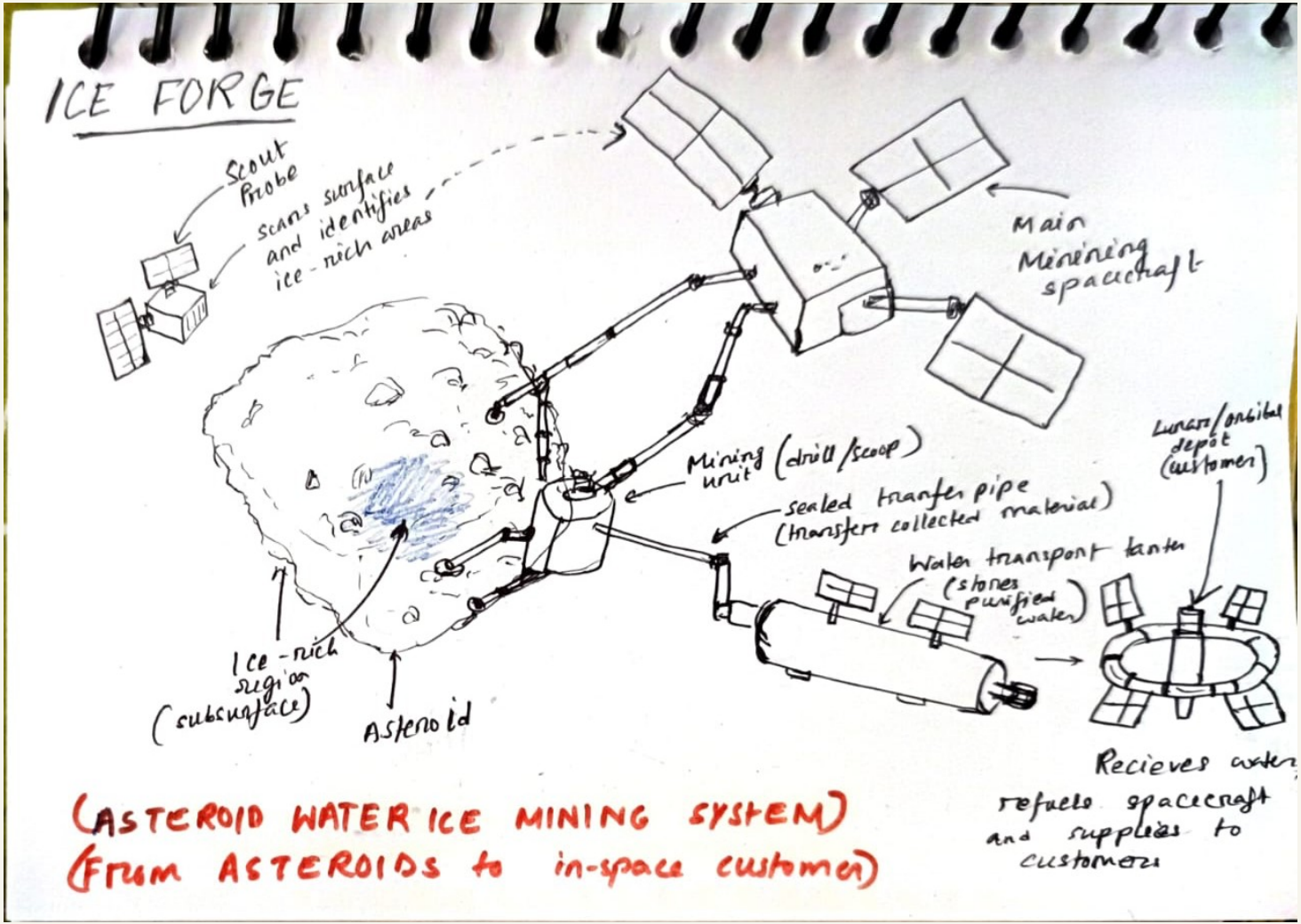
Innovation

Sealed batch cells: material is loaded, closed, heated and condensed without ever exposing dust or vapour to open vacuum.

Economic hypothesis

If risk-adjusted delivered cost stays below about 60% of Earth-launched supply to the same depot, the project is worth scaling.

Evidence labels used throughout: A verified · B derived · C engineering assumption · D illustrative · E unknown. No source was opened during this build; see notes.


ICE FORGE
Scout Probe
scans surface and identifies ice-rich areas
Main Mining spacecraft
Mining unit (drill/scoop)
sealed transfer pipe (transfers collected material)
Water transport tanker (stores purified water)
Lunar/orbital depot (customer)
Ice-rich region (subsurface)
Asteroid
(ASTEROID WATER ICE MINING SYSTEM)
(FROM ASTEROIDS to in-space customer)
Recieves water
refuels spacecraft and supplies to customers

3 Technical design and bottleneck

Bottleneck: in microgravity, loose dust and vapour escape and heat cannot be reused

Baseline mass and energy balance per hour

Step	Value	Basis
Regolith fed	100 kg/h	C
Water present (5 wt%)	5.0 kg/h	D
Extracted (90%)	4.5 kg/h	C
Captured (90%)	4.05 kg/h	C
Usable after purify (98%)	3.97 kg/h	C
Energy per kg water	≈ 4.0 MJ	B
Average / peak power	7 / 11 kW	C

Energy: ice 150→273 K ≈ 0.25 MJ/kg + sublimation ≈ 2.83 MJ/kg (A, to cite) + 9 kg regolith × 0.8 kJ/kg·K × 150 K ≈ 1.1 MJ, 60% of that recovered = 0.85 heater efficiency.

Innovation: sealed batch cells with cascaded heat reuse

Problem: open hoppers lose dust and vapour; hot tailings carry away heat.

Mechanism: scoop fills a cell, lid seals, a heater drives vapour to the cold trap, a hot spent cell then warms the next one by contact plates.

Metric: capture ≥ 90% and ≤ 4.5 MJ/kg. Cost: ~60 kg and more moving parts.

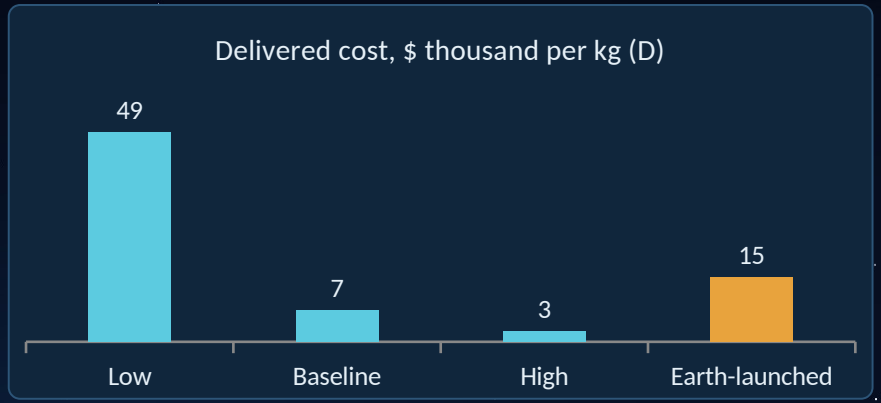
Falsifying test: thermal-vacuum run with ice-regolith simulant; if capture < 75% or leaks exceed budget, switch to the simple open oven. Status: proposed integration of known parts.

Failure mode	Detection	Mitigation	Residual
Anchor slips in low gravity	force + IMU drift	tether; thrusters hold off; abort to hover	Medium
Cell lid seal leaks	pressure + mass-flow check	double lid; reject leaking cell	Low
Ice plugs condenser or valve	pressure rise	two parallel traps, heated bypass	Low–Med
Link delay or fault	heartbeat loss	safe mode: sun-point, heaters off, wait	Low

Ice fraction 5% is illustrative (D); Bennu and Ryugu sampling showed hydrated minerals exist, not ice concentration at a target. Missing data: ice fraction, depth, strength.

4 Cost and conventional comparison

Cost per kg = lifecycle cost ÷ usable water delivered, same destination, same boundary



Baseline: $400 M ÷ (73 t × 75% mission success) = $7.3k/kg. Low: 16 t, 50% success. High: 185 t, 85% success. Earth-launched to cislunar: $15k/kg (D, to verify).

Baseline $400 M (D)	$M
Prospecting probe	40
Tug + mining hardware	120
Launch + transport	90
Operations + comms	50
Spares + contingency	70
Add-l testing	30

All dollar values are illustrative scenario inputs (D), not market facts. Assumes reusable launch and a solar-electric tug exist (technology assumption).

1 Launch and transport (22%)

Mass to orbit sets the bill. Cut by a small probe first, modular payloads and a reusable tug.

2 Energy and processing (≈ 30%)

Array and cell mass scale with kg of water per hour. Cut by heat reuse and processing only confirmed ice-bearing material.

3 Maintenance and failure risk (≈ 25%)

A dead machine ends revenue. Cut by swappable modules, fault detection, and staged tests.

Conventional alternative: launch water from Earth to the same depot. Below about 40 t of delivered water the baseline loses to it; the whole case rests on ice fraction, uptime and success probability.

5 Customer and business model

Product: purified water delivered to a cislunar depot, priced under the customer's alternative

Customer	Buys	Alternative today	Evidence needed
Depot / tug operators	water → propellant, ≥ 10 t lots	Earth launch	signed offtake
Station operators	life-support top-up	Earth resupply	letter of intent
Space agencies	anchor purchase, demo tests	own supply	funded contract

Destination	For	Against
Return to Earth	known market	water is cheap on Earth; large return mass
Use in space (ours)	no Earth round trip	demand unproven
Sell elsewhere in space	flexible buyers	extra transfer legs

Revenue model

Price $8k/kg (D) under the $15k/kg alternative. If it succeeds: 73 t × $8k = $584 M revenue vs $400 M cost. Risk-adjusted: $438 M expected, so margin is thin. Revenue is not profit: pay for operations first, then recover capital.

Risk	Consequence	Mitigation	Warning sign
Technical: low ice fraction	cost doubles	measure before mining	< 3 wt% at prospect
Market: no buyer	no revenue	signed offtake before build	no LOI by gate 3
Financial: capital cost overrun	project stalls	stage funding by gates	> 20% over budget
Operational: failure far away	mission ends	modular spares, safe modes	repeat faults

Demand is a hypothesis. Verified demand = signed commitment with price, volume, quality and delivery location.

6 Feasibility, roadmap and investment decision

Fund stage by stage; each gate can stop, redesign or scale the project

Scenario	Ice	Uptime	Output (3 yr)	Success	$/kg
Low	2%	50%	16 t	50%	48.8k
Baseline	5%	70%	73 t	75%	7.3k
High	10%	85%	185 t	85%	2.5k

Sensitivity (baseline $7.3k/kg)	Change
Ice fraction halves (2.5%)	→ $14.6k
Success 75% → 50%	→ $11k
Life 3 → 2 years	→ ~$11k
Transport +50%	→ $8.1k

1 Lab
cells and trap tested in vacuum chamber

2 Prospect
measured ice ≥ 3 wt%, reachable

3 Demo
extract 100 kg; capture ≥ 85%

4 Quality
water meets customer spec

5 Market
signed offtake; price < 60% of alt.

6 Scale
repeat only if all gates pass

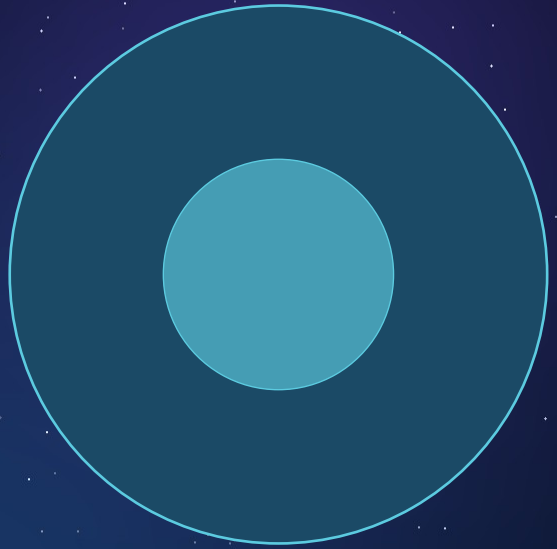
Strongest reason to pursue
A customer already in space avoids the most expensive leg, Earth launch, and staging limits the money at risk before ice is confirmed.

Strongest reason it might fail
Unknown ice concentration and no signed buyer. At low output the delivered cost is several times the alternative; stop if gate 2 or 5 fails.

Scenario values are illustrative (D). Gate thresholds are our proposals, not brief requirements.

Thank you

Water ice is worth mining only if the delivered cost beats Earth supply. We would rather prove that in stages than assume it.



Questions and challenges welcome.